Can disease spread be modelled by a reaction-diffusion equation?

Oliver Clift[1], James Emberton[1], Isaac Millen[1], Nicholas Mitchiner[1], Edward Phillips[1] and Dominik Szablonski[1]

[1]Department of Physics, University of Manchester

GROUP 10

Abstract

In this study, we modelled the spread of disease in Manchester through a reaction-diffusion equation, comparing this to pre-lockdown Covid data. The main assumptions of the model were a constant population density, a recovery time of 14 days, and effects on immunocompromised people is ignored. The Euler method is used to solve the equation computationally. Our study successfully/unsuccessfully modelled spatial disease spread when comparing to pre-pandemic Covid data, but only the initial spread. The model could be improved by implementing random-walks to model more random behaviour, modelling more outbreak centres, or accounting for disease spread reduction.

Introduction

Modelling disease spread is important for governments have appropriate plans in case of large-scale pandemics. The recent Covid pandemic was not handled very well (give some examples, probably cite some things here). With this motivation, we attempted to model the disease spread within greater Manchester. Using our knowledge of physics, we supposed that the spread of a disease like Covid could be modelled by a diffusion equation.

However, due to mass conservation, a standard diffusion equation is incapable of accurately modelling disease spread, because diseases are inherently create new infections. Therefore, for disease modelling, a reaction-diffusion equation is much so that the number of infections at any time is not conserved. This allows us to model new infections and patient recovery.

Theory - Reaction-Diffusion Equation

We attempt to model the spread of disease via a reaction-diffusion equation,

$$\frac{\partial I}{\partial t} = D\nabla^2 I + \beta(x,y)SI - \gamma I, \tag{1}$$

which models spatial densities ρ, which are parametrised such that,

$$\rho = S + I + R, \tag{2}$$

where,

- $S(x,y,t)$: Susceptible population.
  - Those who can catch disease.
- $I(x,y,t)$: Infected population.
  - Those who have the disease.
- $R(x,y,t)$: Recovered population.
  - Those who are no longer infected.

Results: Simulation vs Data

Put things here

Results: Figures

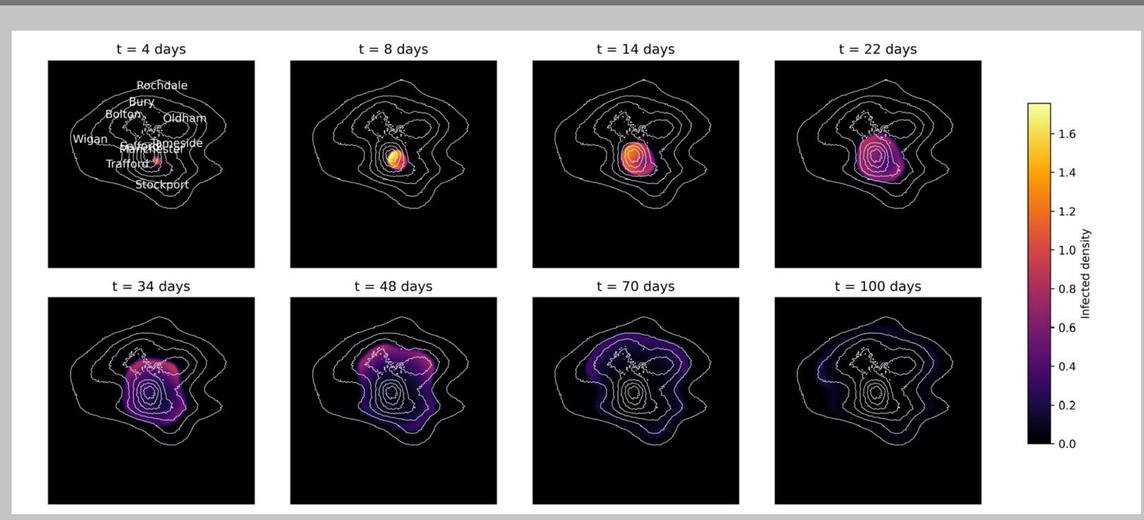

Figure 1: Disease spreading over time in Greater Manchester.

Discussion

We compared our model's output to that of the Covid pandemic. We founf that though our model was successful at modelling the initial spread of the infection, our results started to diverge as time went on. This is likely due to the assumption made that infected people do not change their behaviour when infected. This isn't true in real life, as infected people started to self isolate, and the country went into lockdown in March of 2020. This is a limitation of the model, but suggests that the model would be successful in modelling the behaviour of a pandemic should there be no outside intervention.

Conclusions

We things out ig.

References